Memorandum

To: Pilot Plant Group
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Subject: CHP Model and Modifications

The Cornell Heating and Power Plant is analyzed utilizing ASPEN visualization and optimization software, deriving modifications to the model to yield 12.1% increase in plant efficiency.

Executive Summary

This memo summarizes the results and insights from our analytical model of the Cornell Combined Heat and Power Plant (CHP). Our primary objective is to vary parameters within the Brayton and/or Rankine cycles and evaluate their impacts on overall system efficiency. To assess these impacts, we compare the performance of our modified cycle configurations against current CHP operating data. Our goal is to identify potential improvements in electricity generation, profitability, carbon footprint, and other relevant performance metrics.

In addition to the quantitative analysis, we also provide qualitative interpretations to explain the drivers behind changes in efficiency and why these changes matter in a real-world context. Finally, we outline a practical plan of action for implementing the most promising modifications within the CHP system and propose several additional improvement ideas that extend beyond the scope of our analytical model.

We observed that increasing inlet temperature and pressure to optimized values of 42.22 ^{0}C and 1419 kPa provided an increased efficiency of 12.1%, greatly reducing the cost of operation and energy consumption of the plant. These improvements are conceptualized in various plant running conditions and are followed by a discussion of various improvements to fuel sourcing to the plant. Explicitly, we explore the application of local cattle ruminants as a bio-available energy source for both a contingency plan and additional supply.

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Introduction

Cornell's Combined Heat and Power Plant (CHP) integrates heating, cooling, and electricity generation across its 2,300-acre campus in Ithaca, NY. The facility primarily relies on natural gas—composed mainly of methane along with other hydrocarbons—as its energy source. This natural gas is supplied through several channels, including the NYSEG grid and potential on-campus or renewable sources.

The CHP utilizes two distinct Brayton cycles operating with air as the working fluid. In each cycle, air is compressed, mixed with methane in a combustor, passed through a heat exchanger, and then expanded through a turbine. The exhaust from these Brayton cycles feeds into a subsequent Rankine cycle to further recover energy and maximize overall efficiency. Together, these integrated cycles supply electricity through thermal energy for campus operations.

Using ASPEN, we developed an analytical model that captures the major inputs, outputs, and energy flows present within these cycles. While the model approximates the plant's operation, it does not account for every complexity of the CHP plant. Our goal is to identify opportunities for improvement by analyzing and testing potential modifications within this analytical framework.

Methods

The Cornell Combined Heat and Power Plant runs on a Brayton and Rankine Cycle. The Brayton Cycle extracts work from the combustion of air and natural gas, and the Rankine Cycle uses the expansion of superheated steam to generate power. These two cycles are connected by an HRSG (heat recovery steam generator) system, comprised of three heat exchangers which cool the hot exhaust gases from the Brayton Cycle to produce superheated steam for the Rankine Cycle. Despite their differences, utilizing both the Brayton and Rankine Cycles maximizes power output and efficiency for campus heating and cooling operations.

At the start of the Brayton Cycle, air (st 1) is passed through a compressor (unit K-100). The effluent compressed air (st 2) enters a combustion chamber (CRV-100) with natural gas (st 3). In our analysis, we assumed the natural gas is 100% methane (CH_4). Air is in excess, so 100% of the methane is combusted according to the reaction $CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$. The liquid effluent (st 5) exits the cycle. The vapor portion of the combustion products (st 4) enters a

turbine (K-101) and expands, producing work. In our model, the product of the combustion reaction was 100% vapor, resulting in a negligible flow in the liquid stream (st 5). Exhaust gases (st 6) leaving the turbine enter the HRSG system, composed of units E-100 (superheater), E-101 (evaporator), and E-102 (economizer). The feedwater (st 10) is heated to its saturation temperature at 2930 kPa (st 11), fully vaporized in the evaporator, and then the resulting steam (st 12) is superheated before entering the turbine. Throughout our simulation, we set a zero pressure drop for all heat exchangers and condensers. In other words, friction was neglected such that all streams passing through a heat exchanger in the same direction have the same pressure. The HRSG system exhibits countercurrent flow. The hot exhaust gas (sts 6,7,8) is successively cooled as it travels down the exchangers, while water (sts 11, 12, 13) is heated as it travels up.

Superheated steam leaving superheater E-100 (st 13) is expanded through turbine K-102, producing work. It leaves the turbine as saturated steam (st 14). The saturated steam enters condenser E-103 and leaves as saturated liquid water (st 15), producing thermal energy. Stream 15 is compressed through pump P-100, and the high pressure liquid (st 16) is recycled back into the economizer, where it will be successively heated to superheated steam in the HRSG system.

The energy inputs and outputs in the Brayton Cycle occur at the compressor and turbine (K-100 and K-101). Work is required at the compressor, while work is extracted at the turbine. For the Rankine Cycle, work is input and output during the water compression through the pump and superheated steam expansion through the turbine, respectively. Heat is extracted (Q out) during the condensation of the saturated steam (unit E-103).

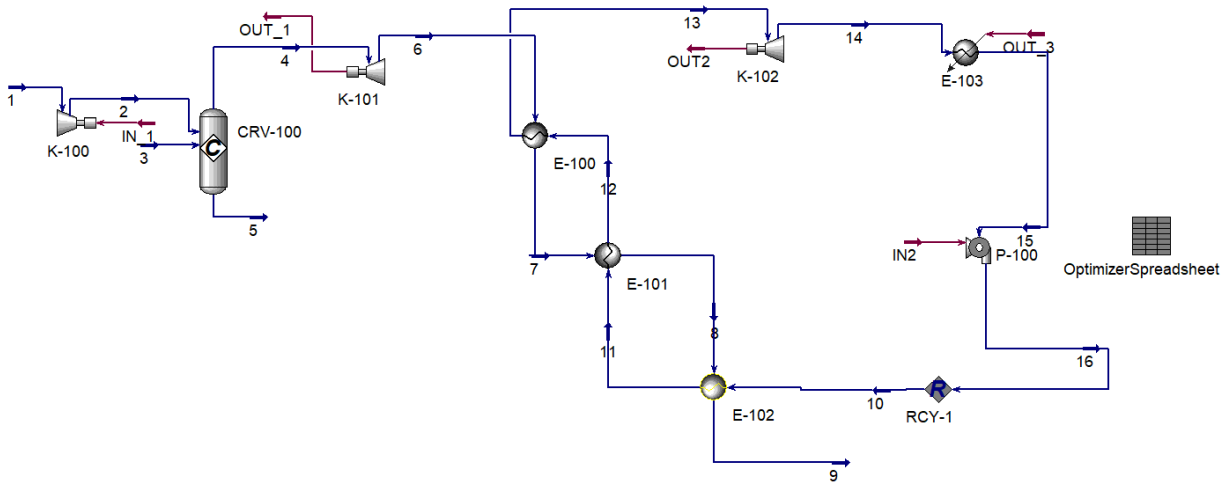


Figure 1. Our ASPEN simulation process flow diagram.

In this process, we sought to increase overall efficiency by adjusting the conditions of the inlet air and gas streams. This approach preserves consistent operation of existing units without requiring costly modifications to the machinery. In our analysis, we set the inlet pressure, temperature, and mass flow rates as variables within our system. We then set the optimizer function to maximize the efficiency equation input as:

$$\eta = \frac{Wout - Win}{Qin} = \frac{(Out1 + Out2) - (In1 + In2)}{m_{gas} * \Delta H_{combustion}} \text{ (Eqn. 1)}$$

η is the overall efficiency of the process, W_{out} and W_{in} is the overall work input and output for the system. Q_{in} is the heat input at the combustion stage, defined by m_{gas}, or the flow rate of natural gas, and $\Delta H_{combustion}$ = - 55500 kJ/kg [1] which is the heat of combustion of natural gas.

We varied the stated parameters within the inlet streams to determine the most optimal improvements.

Results and Discussion

As discussed, we decided to use the optimizer function within ASPEN to change the characteristics of the air and methane inlet streams to optimize the efficiency of the process. To calculate the efficiency of the process, the following equation was used:

$$\eta = \frac{Wout - Win}{Qin} = \frac{(Out1 + Out2) - (In1 + In2)}{m_{gas} * \Delta H_{combustion}} \text{ (Eqn. 1)}$$

Table 2 details the optimized values for mass flow rate, temperature, and pressure of the air and gas inlet streams.

Parameter	Air Initial	Air Final	Gas Initial	Gas Final
Temperature (^{0}C)	21.11	42.22	42.78	56.68
Pressure (kPa)	709.5	1419	3075	3018
Mass Flow (kg/h)	4.990 x 10^4	4.995 x 10^4	2000	1000

With these altered values, efficiency increases from 38.4% to 50.5%.

These changes do bring about a higher efficiency in the CHP plant, but there are some other aspects to consider when analyzing this efficiency change such as costs, energy output, feasibility, sustainability, etc. One of the first things to analyze is the energy output at this higher efficiency. Table 3 provides a summary of the changes in energy output:

Output	Before Change (kJ/h)	After Change (kJ/h)	% Change
Out_1	4.353 x 10^7	2.535 x 10^7	-42%
Out_2	2.947 x 10^6	2.947 x 10^6	0%

With this change in energy output, there is a 39% overall decrease in energy output. Therefore this solution should be approached with some caution in implementation. If energy demand is at a high, this may not be feasible, but during the off/summer months, this may be an efficient solution.

Another thing to consider is the cost reduction that comes with this change. There are two aspects of cost that may change substantially, that being raw material cost of the gas, as well as the change in input energy. For the raw material input, there will be a 50% decrease in total cost,

from 2,520 USD/h to 1,260 USD/h (with gas priced at 1.26 USD/kg) [2]. This is annualized to a cost reduction of approximately 11 million dollars (assuming constant operation for every hour of the year). The calculations for the raw material cost reduction are as follows:

$$\text{Cost Reduction} = m_{gas} * price = (\frac{1000kg}{h})(\frac{1.26\$}{kg}) = 1260\ USD/h \text{ (Eqn. 2)}$$

$$(\frac{1260\$}{h})(\frac{24h}{d})(\frac{365d}{yr}) = 11.04\ million\ USD/yr$$

Cost Reduction due to raw material input is defined as a function of inlet natural gas mass flow rate and the price of natural gas.

For the decrease in energy input we must first analyze this change in Table 4:

Input	Before Change (kJ/h)	After Change (kJ/h)	% Change
In_1	4.227×10^6	2.632×10^5	-93%
In_2	2.506×10^4	2.506×10^4	0%

With the observed change in energy, there is a 93% total decrease in energy input, correlating to a 93% decrease in energy input cost. This decrease is from 283 USD/h to 19.22 USD/h (using a cost of .24 USD/kWh) [3]. This is annualized to a cost reduction of 2.3 million dollars (assuming constant operation for every hour of the year). In totality, we see that with this optimization, there is a 54% reduction in total cost, totalling approximately 13.3 million dollars per year. The calculations for the energy input cost reduction is as follows:

$$Cost = (\frac{In1+In2}{3600}) * price = (\frac{2.632x10^5+2.506x10^4}{3600}kWh)(\frac{.24USD}{kWh}) = 19.22\ USD/h \text{ (Eqn. 3)}$$

$$19.22\$/h * 24h/d * 365d/yr = 168,367\ USD/yr$$

Cost Reduction due to decreased energy input is defined by the required input energy for the compressors in the Brayton (In1) and Rankine (In2) Cycles and the price of energy at 0.24 USD/kWh.

Lastly, there are aspects to consider surrounding the feasibility of these changes. For example, raising the inlet temperature from 21.11 to 42.22 ^{0}C would require energy upon entry given that the original temperature entered at room temperature. Another aspect that may not be feasible would be raising the pressure to 1409 kPa. This is a pressure significantly higher than the original, but also much higher than atmospheric pressure. This too, is not represented in the overall energy balance of our process. These parameters are not infeasible to reach, however, they will require work that is not necessarily accounted for in our preliminary model. Our model also failed to account for the vaporization of both high pressure and low pressure steam in the HRSG. While ASPEN modeling limited our ability to include this unit, we believe that the inclusion of this high pressure steam loop would not influence our optimization findings. However, it is highly feasible that both the existing and suggested system would have higher observed efficiencies due to this system.

In addition to direct improvements to the Brayton and Rankine process, we wanted to explore alternative improvements to the CHP fuel sourcing and operation. The current contingency plan for the plant's NYSEG natural gas supply relies on a 1,000 L supply tank of liquid methane. Assuming a density of liquid methane that corresponds to roughly 932 lb of stored natural gas (Eqn. 4), this backup would only support full campus energy production for about 9 minutes (Eqn. 5), given the plant's nominal fuel consumption of ~6,000 lb/h or ~54,000,000 lb/yr (Eqn. 6).

$$(1000\ L)\left(\frac{422\ kg}{1000\ L}\right)\left(\frac{2.12\ lb}{1\ kg}\right) = 932\ lb\ Natural\ Gas \text{ (Eqn. 4)}$$

$$\text{Time} = 932\ lb\left(\frac{6150\ lb}{h}\right)\left(\frac{60\ m}{1\ h}\right) = 9.22\ minutes \text{ (Eqn. 5)}$$

$$m_{gas} = \left(\frac{6150\ lb}{h}\right)\left(\frac{24\ h}{1\ d}\right)\left(\frac{365\ d}{1\ yr}\right) = 53,874,000\ lb\ Natural\ Gas/yr \text{ (Eqn. 6)}$$

Over a 12-hour outage, this storage would therefore be capable of supplying only about 1.26% of the fuel required for continuous full-load operation (Eqn. 7).

$$Provided\ Fractional\ Power = \left(\frac{932\ lb}{12\ h\left(\frac{6150\ lb}{h}\right)}\right) = 0.0126 \text{ (Eqn. 7)}$$

The reported fractional power over a 12-hour period represents the portion of total energy production that this contingency supply can sustain relative to the plant's typical demand.

Ideally, this reservoir would be used solely as startup fuel to initiate the Brayton cycle and transition the plant back to internal energy production. However, in the event of an extended interruption in the NYSEG supply, this internal energy production would prove an insufficient load for campus heating and power production. Natural methane reservoirs exist for the continuous production and storage of methane for combustion or alternative energy production.

At the Cornell University Ruminant Center (CURC), there are approximately 700 adult cows and 420 youngstock [4]. An adult cow produces roughly 220 lb of methane per year through enteric emissions, which can be obtained with manure management [5]. Assuming proportionally lower emissions from youngstock, this 2,000-acre farm could generate on the order of 225,000 lb of methane annually (Eqns. 8, 9, & 10). With the implementation of various venting, extraction, and purification processes to capture and upgrade this sourced, bioavailable methane, CURC could function as a modest, on-site renewable fuel source for the campus CHP plant.

$$Adult\ Cow\ Methane\ Produced\ = 700\ Adult\ Cows\ yr\left(\frac{220\ lb}{yr}\right) = 154,000\ lb/yr \quad \text{(Eqn. 8)}$$

$$Youngstock\ Cow\ Methane\ Produced\ = 420\ Youngstock\ Cows\ yr\left(\frac{180\ lb}{yr}\right) = 75,600\ lb/yr \quad \text{(Eqn. 9)}$$

$$Total\ Methane\ Produced\ = 154,000\ lb/yr\ +\ 75,600\ lb/yr\ = 224,600\ lb/yr \quad \text{(Eqn. 10)}$$

Each year of living, an adult and youngstock cow produces 220 and 180 lb of methane/yr respectively. The manure of these cows holds a quantifiable total methane production. The total methane produced by both adult and youngstock cows is the sum of totals produced by each type of cow.

In the local Tompkins County area, there were an estimated 26,000 cattle in the 2022 census [6]. Even assuming a 100% adoption rate of this technology and extensive (and expensive) piping infrastructure, this would only provide approximately 10% of the natural-gas requirement for campus energy production (Eqn. 11 & 12).

$$26,000\ adult\ cattle\left(\frac{220\ lb}{year}\right) = 5,700,000\ lb\ Methane/yr \text{ (Eqn. 11)}$$

$$Materials\ Fraction = \frac{Methane\ Available\ from\ Cows}{Natural\ Gas\ Requirement} = \frac{5{,}700{,}000\ lb/yr}{53{,}874{,}000\ lb/yr} = 10.6\% \text{ (Eqn. 12)}$$

The estimated number of adult cattle and previously stated methane production rate of adult cows was used to determine the potential annual production of methane from Ithaca cattle.The Materials Fraction compares the percent of methane accessible from 26,000 local Tompkins County Cattle to the CHP Natural Gas Requirement.

The implementation and management costs of these technologies would far exceed the $1,100,000 USD annual savings, making a complete replacement of the NYSEG supply infeasible despite the clear environmental advantages.

However, we believe that small-scale natural gas reservoirs near campus could prove highly valuable with respect to contingency-plan methane supply, greenhouse-gas reduction, and future fuel-cell energy production in the region. The existing liquid-methane supply tank provides only 1.26% of full campus power over a 12-hour period before being depleted. In contrast, the bioavailable, methane-trapping capacity of cattle ruminance offers a continuous source of methane, providing ~0.5% of the fuel required for full-power operation per hour. When accounting for potential storage strategies and the need for long-duration contingency capability, methane purification from local agricultural sources represents a reasonable supplement for NYSEG outages rather than a full replacement.

Several green-energy companies, such as Bloom Energy, regularly partner with individual cattle farms to harvest methane emissions. Bloom Energy fuel-cell systems convert methane into electricity by electrochemically oxidizing methane. This technology performs a non-combustion reaction where it splits methane into hydrogen and carbon dioxide, before producing power in

the solid-oxide cell stack [7]. This application of renewable energy production could be implemented at CURC and other Tompkins County farms. Introducing methane extraction and fuel-cell technologies to the local agricultural sector could play a significant role in advancing regional green-energy initiatives. Typically, a cattle farm can produce enough methane to meet much of its own electricity demand using Bloom, or similar, local fuel-cell systems. This could help both Cornell and the local communities reach the 2035 Energy Conservation Initiative [8].

Conclusion

Based upon our model and the optimization we ran, the results gave an increased efficiency of 12.1%. This increased efficiency came with a decrease in cost by around 13.3 million dollars per year through reduction in energy consumption and methane input. This increase in efficiency also came with a 39% decrease in overall energy output. In general, we believe a higher inlet temperature and pressure, approaching the optimized values, would drive more efficient operating conditions. With these results, we would recommend going through with these corrections, but potentially not all of them during all operating conditions. For example, we recommend varying the methane flow rate, however not making it this minimum during all operating time. Instead, vary it with the energy demand that is present within campus. Another aspect to go through with, would be varying the pressure in the direction of the optimized value, so potentially not going all the way to 1419 kPa, rather move it towards a value of 1000 kPa. In addition to direct process optimization, we suggest exploring the impact and application of natural methane sources such as cattle ruminance. These biogas recovery systems would prove an excellence contingency plan and be influential in spreading green-energy in the Tompkins County area.

References

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Appendix

Table 1 - Operating Conditions of ASPEN Model Streams

Stream	Temperature (^{0}C)	Pressure (kPa)	Mass Flow Rate (kg/h)
1	42.22	1419	4.995×10^4
2	47.47	1503	4.995×10^4
3	56.68	3018	1000
4	895.5	1503	5.095×10^4
5	895.5	1503	0 (negligible)
6	481.1	110	5.095×10^4
7	461.7	110	5.095×10^4
8	260.4	110	5.095×10^4
9	192.5	110	5.095×10^4
10	102.7	2930	6304
11	231.1	2930	6304
12	233.3	2930	6304
13	315	2930	6304
14	102.3	110	6304
15	102.3	110	6304
16	102.7	2930	6304